02 Acids and Bases Problem-solving approach

Basis: `evidence/che/notes/AB.txt` (AB1–AB9, 157 sub-question annotations in total: 49 Section One multiple-choice questions and 108 in Section Two and Section Three, the latter listing the question stem, key and exam report for each sub-question), `tools/parttags_CHE.txt` (sub-question tags), `tools/catalogue_CHE.json` (pattern definitions), `evidence/che/notes/INDEX.md` (mark allocation statistics). Resources available to students are taken from `CHE/papers/2025_CHE_DataBooklet.pdf`, with additional spot checks of the 2016, 2019, 2022 and 2024 data booklets. Past-paper question reference format: `2024S-Q30(a)` = 2024 Section Two (short answer) Question 30 (a); `2016M-Q10` = 2016 Section One (multiple choice) Question 10; `2019E-Q37(e)` = Section Three (extended answer). `M`, `S` and `E` indicate the paper section only. ★ marker: ranked by INDEX mark allocation, the top approximately one quarter (9 patterns in this topic, approximately 2): AB9 (71 marks); AB4 and AB5 tied (55 marks each), both marked. Review status: independent review completed (against past exam questions, key and exam report); record at `CHE/methods/审校记录.md`.

First, remember: where markers repeatedly deduct marks

- Calculations must show working and include units: the key for 2016E-Q42(f) notes "No marks awarded unless accompanied by supporting working"; a pH result on its own scores 0.
- "weak" and "dilute" are different: strong versus weak refers to the degree of ionisation, while concentrated versus dilute refers to concentration (2023S-Q34(c) requires using Ka and partial ionisation to explain a "weak acid").
- Explanations of hydrolysis and buffers must include equations: the key repeatedly requires at least one or two equations (2022S-Q28, 2023S-Q28(a), 2024S-Q32(a)).
- Exam report comments on this topic: the 2022 report lists "explaining the difference between solutions formed by two salts" (2022S-Q28, i.e. AB3) as one of the most difficult questions; the 2023 report lists 2023S-Q26 (concepts and equations of acid–base theory) as one of the best-answered questions, and lists 2023S-Q28 (definition and explanation of buffer systems) and 2023S-Q32 (intermolecular forces and solubility) as the most difficult questions; the 2022 report states that titration calculations and practical technique (2022E-Q38) were challenging for many candidates.
- Do not confuse Arrhenius and Brønsted–Lowry: the former defines an acid as producing H^+ (or H_3O^+) in water, while the latter defines it by proton transfer (2023S-Q26(d)).

What the data booklet provides (resources available to students in the exam)

- $K_w = 1.00 \times 10^{-14}$ (25 °C); $pH = -\log_{10}[H^+]$ (given in the data booklet).
- Solubility rules (used in AB9 to decide whether a precipitate forms and in AB9 precipitate-weighing questions).
- Colour table (observations in AB9 and AB6): Cu^{2+} blue, CuO black, $CuCO_3$ green (data booklet solids table); the data booklet does not list the colour of $Cu(OH)_2$, so relevant observations are as given in the question stem or key.
- There is no Ka table, and no indicator table; Ka and indicator colour-change ranges are given in the question stem (e.g. 2023S-Q34(c) gives $K_a = 1.73 \times 10^{-5}$; the indicator tables for 2016S-Q34(b) and 2021S-Q28(a) are in the question stem).
- Values of Kw that vary with temperature are given in the question stem (e.g. 2017S-Q31(d) gives $K_w = 5.48 \times 10^{-14}$ at 50 °C).

Pattern 1: Brønsted–Lowry acids and bases and conjugate acid–base pairs

Identification: "Brønsted–Lowry" "conjugate acid–base pair" "amphoteric" "circle the Brønsted–Lowry bases" "label the weak acid and its conjugate base".

Steps

1. Identify the transfer of protons (H^+): the species that donates H^+ is the acid (proton donor), and the species that accepts H^+ is the base (proton acceptor).
2. Pair them: after an acid loses H^+ it becomes the conjugate base; after a base gains H^+ it becomes the

conjugate acid.

3. Amphoteric species (e.g. H_2O, HCO_3^-, $HAsO_4^{2-}$) can act as either an acid or a base: write two equations separately, each showing it acting as an acid or a base (2018S-Q31(bi)).
4. In equations, use ⇌ and label the acid/base role of each species.
5. When explaining solution pH (e.g. why an NH_3 solution is basic), write the equation for NH_3 accepting H^+ and compare $[OH^-]$ with $[H^+]$ (2016S-Q31(b)).

Mark-earning points (typical key wording): ✓ 2016S-Q31(b) (3 marks): states that water acts as an acid and donates a proton, and ammonia acts as a base and accepts a proton (or labels the conjugate acid–base pairs); writes $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$; states that $[OH^-] > [H^+]$, so the solution is basic. ✓ 2019E-Q36(a) (2 marks): correctly labels the acid and base in two conjugate acid–base pairs (H_3O^+ is the acid, H_2O is the conjugate base; HCO_3^- is the base, H_2CO_3 is the conjugate acid). ✓ 2024S-Q29(a) (4 marks combined with PRO4): describes the structure of a zwitterion (the carboxylic acid donates H^+, which is accepted by the amino group, forming a dipolar ion with positive and negative charges). ✓ 2024S-Q32(b) (1 mark): labels NH_4^+ as a weak acid and NH_3 as the conjugate base.

Common errors (exam report): the 2023 report lists 2023S-Q26 as one of the better-answered questions, but marks are still deducted for details. The following common errors are summarised from the key:

- Writing the charge of the conjugate base incorrectly (after removing H^+, the charge should decrease by 1).
- Reversing the acid and base roles (H_3O^+ is an acid, not a base).
- Writing only one equation for an amphoteric species (2018S-Q31(bi) requires two equations).
- Missing one of the B-L bases when circling bases (the key for 2023S-Q26(c) lists three: H_2O, NO_2^-, $HCOO^-$; all must be circled for full marks).
- MC questions ask for conjugate acid–base pairs or whether a species acts as an acid or a base (e.g. 2016M-Q10, 2024M-Q3, 2025M-Q25); first mark the proton transferred, then check which species gains or loses H^+.

Past questions: 2016M-Q10, 2016M-Q11, 2016S-Q31(b), 2017S-Q31(a), 2017E-Q40(f), 2018M-Q8, 2018S-Q31(b)(i), 2019M-Q24, 2019E-Q36(a), 2020M-Q2, 2020M-Q15, 2021M-Q21, 2022M-Q14, 2023S-Q26(b), 2023S-Q26(c), 2024M-Q3, 2024S-Q29(a), 2024S-Q32(b), 2025M-Q20, 2025M-Q25

Pattern 2: Strong/weak acids and bases, ionisation and Ka; species in solution

Recognition: "write the ionisation equation" "polyprotic / monoprotic" "Ka" "list the species present in solution" "percentage ionised" "Arrhenius vs Brønsted–Lowry" .

Steps

1. Distinguish strong/weak from concentrated/dilute: strong/weak refers to the degree of ionisation, concentrated/dilute refers to concentration. For weak acid questions, state "partial / incomplete ionisation" (2023S-Q34(c)).
2. For polyprotic acids, write stepwise ionisation; each step loses only one H^+ and uses ⇌; the product of the first step is the reactant of the second (e.g. $H_2PO_4^-$ is both the product of the first ionisation of H_3PO_4 and the reactant of the second; 2018S-Q27(a): $H_2PO_4^- + H_2O \rightleftharpoons HPO_4^{2-} + H_3O^+$).
3. For stepwise ionisation structure questions (e.g. citric acid, 2016S-Q33(b)): for each step, draw the H^+ removed; the negative charge must appear on the deprotonated oxygen atom.
4. List the main species in solution: solvent H_2O; weak acid molecules and their ions; H_3O^+ and OH^- from the self-ionisation of H_2O; dissolved gases and their derivatives (e.g. dissolved CO_2 forms H_2CO_3, HCO_3^-, CO_3^{2-}, see 2018S-Q27(b)).
5. Compare acid strength: the larger the Ka, the stronger the acid. Ka < 1 means reactants predominate (2023S-Q34(c)).
6. From pH find $[H_3O^+]$, then find percentage ionisation: $[H_3O^+] = 10^{-pH}$; percentage = amount in mol of acid ionised ÷ original amount in mol of acid × 100% (2023S-Q34(b): 2.36%).
7. Arrhenius and Brønsted–Lowry: an Arrhenius acid produces H^+ in water; a Brønsted–Lowry acid is a

proton donor (2023S-Q26(d), the Arrhenius equation must be written).

Scoring points (typical wording in the key): ✓ 2023S-Q26(a) (4 marks): monoprotic means one ionisable H^+, polyprotic means two or more (2 marks); give two suitable equations (2 marks). ✓ 2023S-Q34(b) (4 marks): $[H_3O^+] = 10^{-2.23} = 5.89 \times 10^{-3}$ mol L^{-1}; $n(H_3O^+)$ = n(acid ionised); percentage = $5.89 \times 10^{-3} \div 0.250 \times 100$ = 2.36%. ✓ 2023S-Q34(c) (3 marks): weak acid undergoes partial (incomplete) ionisation; the percentage ionisation in (b) is very small; Ka < 1 means reactants predominate. ✓ 2016S-Q33(b) (3 marks): 1 mark per step of ionisation; the negative charge must be on the correct oxygen atom. ✓ 2018S-Q27(a) (2 marks): second-step ionisation equation (one minor error allowed, key notes "one minor error"). ✓ 2018S-Q27(b) (3 marks): key only counts the first three items (e.g. $MgCO_3(s)$, $Mg_3(PO_4)_2(s)$, $H_2CO_3(aq)$, etc.). ✓ 2020S-Q31(a) (3 marks): write in order $CO_2(g) + H_2O(\ell) \rightleftharpoons H_2CO_3(aq)$, $H_2CO_3(aq) + H_2O(\ell) \rightleftharpoons HCO_3^-(aq) + H_3O^+(aq)$, $HCO_3^-(aq) + H_2O(\ell) \rightleftharpoons CO_3^{2-}(aq) + H_3O^+(aq)$ (1 mark for each species in each box; state symbols not required). ✓ 2016E-Q42(g) (6 marks): find $[H^+]$ from pH, subtract the part provided by CO_2, find the $[H^+]$ provided by SO_2; treat H_2SO_3 as fully ionised, each mole gives 2 mol H^+; then use pV = nRT (T = 289.15 K, P = 97.2 kPa) to find the volume of SO_2.

Common errors (exam reports): The 2023 report lists 2023S-Q26 as a relatively well-answered question ((a) and (d) both belong to this pattern). The following common errors are summarised from the key:

- Writing a polyprotic acid in one step as $H_3PO_4 \rightleftharpoons PO_4^{3-} + 3H^+$.
- Writing a strong acid as reversible (e.g. $HCl \rightleftharpoons H^+ + Cl^-$ using a double arrow).
- Treating "concentrated" as "strong" (2023S-Q34(c) requires a clear distinction).
- Using Ka to express percentage ionisation, or using pH directly instead of Ka.
- When using the Arrhenius definition, writing only "H^+" but not stating that it is produced in aqueous solution (key accepts "produce H^+ in aqueous solution").
- MC questions compare strong and weak acids, the degree of ionisation and Ka, or ask which species are present in solution (e.g. 2018M-Q3, 2024M-Q11, 2025M-Q2); a weak acid is only partly ionised, so compare ion concentrations, not just the acid's name or stated concentration.

Past questions: 2016S-Q31(b), 2016S-Q33(b), 2016E-Q42(g), 2017M-Q1, 2018M-Q3, 2018M-Q18, 2018S-Q27(a), 2018S-Q27(b), 2019M-Q8, 2019S-Q31(c), 2019S-Q35(a)(i), 2019S-Q35(a)(ii), 2020M-Q18, 2020S-Q31(a), 2021M-Q3, 2021M-Q8, 2022M-Q11, 2022M-Q13, 2023S-Q26(a), 2023S-Q26(d), 2023S-Q34(b), 2023S-Q34(c), 2024M-Q9, 2024M-Q10, 2024M-Q11, 2024M-Q12, 2025M-Q2, 2025E-Q36(a), 2025E-Q36(b)

Pattern 3: Hydrolysis of salts and acidity/basicity of solutions

Recognition: "Explain why the solution is acidic / basic" "Explain why $KHSO_3$ gives an acidic solution while $KHCO_3$ gives a basic solution" "write the hydrolysis equation". The question stem usually gives a salt (e.g. KCN, NH_4Cl, $KHCO_3$, $KHSO_3$, Na_3PO_4) or an equivalence-point solution.

Steps

1. Determine the sources of the cation and anion: ions from strong acids or strong bases (K^+, Na^+, Cl^-, NO_3^-) do not hydrolyse and are neutral ions.
2. Weak acid anions (e.g. CH_3COO^-, CN^-, PO_4^{3-}, ClO^-, $RCOO^-$) react with water to produce OH^-, making the solution basic. Weak base cations (e.g. NH_4^+) hydrolyse to produce H_3O^+, making the solution acidic.
3. Amphiprotic ions (HCO_3^-, HSO_3^-, $HAsO_4^{2-}$): write their two competing reactions (one produces H_3O^+, one produces OH^-), compare the extent of the two (relative sizes of K values or Ka and Kb), and the predominant one determines the acidity/basicity of the solution (2018S-Q31(bii); 2022S-Q28).
4. Write the equations and compare $[OH^-]$ with $[H_3O^+]$ to reach a conclusion.
5. In a titration, the acidity/basicity of the equivalence-point solution is determined by salt hydrolysis, so the choice of indicator also depends on this conclusion (see Pattern 6; e.g. in 2020E-Q37(d), hydrolysis of sodium phosphate makes the equivalence point weakly basic, pH about 9).

Scoring points (typical wording in the key): ✓ 2022S-Q28 (7 marks): K^+ is neutral and does not react; HSO_3^- and HCO_3^- each undergo hydrolysis; compare the extent of the two reactions; state that $[OH^-] > [H_3O^+]$ is

basic, and the reverse is acidic; at least two equations. ✓ 2019S-Q27(a) (4 marks): hypochlorite hydrolysis makes OH^- in excess ($[OH^-] > [H^+]$); write the equation (reactants and products both correct and balanced, 2 marks). ✓ 2019S-Q27(b) (3 marks): pH decreases after adding hydrochloric acid; write the equation related to calcium hypochlorite (balanced). ✓ 2017E-Q39(e) (2 marks): $RCOO^-$ is the conjugate base of a weak acid, reacts with water to accept H^+ (hydrolysis) and produce OH^-. ✓ 2018S-Q31(bii) (1 mark): the solution is basic, indicating that the extent of reaction 1 is greater than that of reaction 2 (or $K_1 > K_2$). ✓ 2023S-Q33(b) (5 marks): at the equivalence point NH_3 hydrolyses to produce OH^-, $[OH^-] > [H^+]$, choose phenolphthalein, and write $NH_3(aq) + H_2O(\ell) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$.

Common errors (exam reports): The 2022 report notes that "explaining the difference between solutions formed by two salts" (2022S-Q28) was one of the hardest questions on that paper. The following common errors are summarised from the key:

- Thinking $KHSO_3$ is also basic, or writing only one hydrolysis equation.
- When comparing the extents of two hydrolysis reactions, writing only "produces OH^-" without comparing the extents.
- Mistaking salts of strong acids and strong bases (e.g. KCl, $NaNO_3$) as undergoing hydrolysis.
- Omitting "$\rightleftharpoons$" when writing equations, or writing only one side of the hydrolysis.
- MC questions ask about the acidity or basicity of salt solutions or hydrolysis equations (e.g. 2021M-Q9, 2023M-Q18, 2025M-Q19); salts of a strong acid and a weak base give acidic solutions, salts of a strong base and a weak acid give basic solutions, and for hydrogen salts compare ionisation with hydrolysis.

Past questions: 2016S-Q31(a), 2017E-Q39(e), 2018S-Q27(d), 2018S-Q31(b)(ii), 2019S-Q27(a), 2019S-Q27(b), 2020E-Q37(d), 2021M-Q9, 2022S-Q28, 2023M-Q18, 2023S-Q33(b), 2024M-Q4, 2025M-Q19

Pattern 4: Buffer solutions ★

Identify: "buffer", "buffering capacity", "buffer equilibrium", "explain why the pH changes only slightly when a small amount of acid / base is added". The question stem gives a weak acid and its conjugate base system (e.g. H_2CO_3/HCO_3^- blood buffer, $H_3C_6H_5O_7/H_2C_6H_5O_7^-$ citrate buffer, CH_3COOH/CH_3COO^-, $C_6H_5COOH/C_6H_5COO^-$, $HPO_4^{2-}/H_2PO_4^-$, NH_4^+/NH_3).

Steps

1. Write the equilibrium equation for the buffer pair (using $\rightleftharpoons$): $HA + H_2O \rightleftharpoons H_3O^+ + A^-$ (or its reverse).
2. Adding H^+ (strong acid): A^- reacts with H_3O^+, the equilibrium shifts left, and H_3O^+ is consumed.
3. Adding OH^- (strong base): $HA + OH^- \rightarrow A^- + H_2O$, the equilibrium shifts right, and OH^- is consumed.
4. Explain that the pH change is very small: most of the added H^+ or OH^- is consumed by the buffer pair, $[H_3O^+]$ changes only slightly, so the pH change is small (finish with $pH = -\log[H^+]$).
5. Compare with a non-buffered system: pure water does not have enough weak acid or weak base to react with the added ions, so the added H^+ or OH^- causes the full pH change (2016S-Q32(b)).
6. Buffering capacity: depends on the absolute concentrations of the two components (the higher the concentration, the greater the buffering capacity, 2018S-Q29(e)) and on the relative concentrations of the two components (the key for 2024S-Q32(e) only requires listing this factor; "the closer to 1:1, the better" is additional chemistry knowledge, not in the key). Definition of buffering capacity: the amount of acid or base that can be added before the pH changes by more than one unit (2024S-Q32(d)).

Mark-earning points (typical wording in the key): ✓ 2016S-Q32(a) (2 marks): $H_2CO_3 + H_2O \rightleftharpoons H_3O^+ + HCO_3^-$ (or its reverse). ✓ 2016S-Q32(b) (4 marks): adding HCl increases $[H_3O^+]$; the buffer contains enough HCO_3^- to react with it; the equilibrium shifts left (or the reverse reaction rate increases), bringing $[H_3O^+]$ back down; pure water does not have enough OH^-/weak base to react, so the pH changes significantly. ✓ 2018S-Q29(d) (4 marks): OH^- reacts with citric acid, $H_2C_6H_5O_7^-$ increases and $H_3C_6H_5O_7$ decreases, the equilibrium shifts in the direction that produces citric acid, $[H_3O^+]$ decreases but only slightly; the pure system has no weak acid to react. ✓ 2018S-Q29(e) (3 marks): the concentrations of both components increase, so they can react with more H^+ or OH^-; therefore more acid or base can be added before the pH changes noticeably. ✓ 2021S-Q28(bii) (5 marks): definition of a buffer; the region contains the weak acid C_6H_5COOH and its conjugate

base $C_6H_5COO^-$; write the equation; the added OH^- is consumed by benzoic acid, so the pH does not rise noticeably. ✓ 2023S-Q28(a) (3 marks): definition of a buffer (resists significant pH change); H_2CO_3 reacts with OH^-; HCO_3^- reacts with H^+. ✓ 2024S-Q32(c) (3 marks): strong acid increases $[H_3O^+]$; the buffer equilibrium shifts left to counteract the change; the pH only drops a little. ✓ 2025S-Q29(b) (5 marks): write the equation involving OH^- ($CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$); $[OH^-]$ increases; the equilibrium shifts in the direction that consumes OH^-; at the new equilibrium $[H_3O^+]$ decreases, but only slightly; the pH rises slightly (the fifth point in the key).

Common errors (exam reports): The 2023 report lists "defining and explaining a buffer system" (2023S-Q28) as one of the hardest questions on that paper. The 2024 report lists Question 32 (applications of buffers) as one of the hardest questions in Section Two. The following common errors are summarised from the key:

- Writing "the pH does not change" for a buffer solution, when it should be "the pH changes only slightly".
- Not writing the equation for the reaction of the added H^+ or OH^- with the buffer pair.
- Only saying "buffer resists change" without stating which two species are responsible (2023S-Q28(a) requires stating the role of each species).
- Explaining pure water in terms of a buffer (pure water is not a buffer system, 2016S-Q32(b)).
- Missing the double arrow in an equation or getting one species wrong (2018S-Q29(a) notes "one minor error, include lack of double arrow").
- Confusing buffering capacity with pH, or only saying "the concentration is high" without stating which two components.
- MC question 2022M-Q5 asks which equimolar pairs give an acidic buffer; a buffer needs a weak acid with its conjugate base, and a strong acid with its salt cannot buffer.

Past questions: 2016S-Q32(a), 2016S-Q32(b), 2018S-Q29(a), 2018S-Q29(b), 2018S-Q29(d), 2018S-Q29(e), 2019E-Q36(d), 2019E-Q36(e), 2020S-Q30(b), 2021S-Q28(b)(ii), 2022M-Q5, 2023S-Q28(a), 2023S-Q28(b), 2024S-Q32(a), 2024S-Q32(c), 2024S-Q32(d), 2024S-Q32(e), 2025S-Q29(a), 2025S-Q29(b)

Pattern 5: pH, Kw and strong acid–strong base calculations ★

Identify: "Calculate the pH", "Determine the pH of the final solution", "calculate the concentration of the hydroxide ion", "calculate the $H^+(aq)$ and $OH^-(aq)$ concentrations of pure water at …". The question stem gives Kw values that change with temperature, or gives the volumes and concentrations of two solutions being mixed.

Steps

1. Use n = cV to find the amount in mol of H^+ or OH^- (convert mL to L; $Ca(OH)_2$ and $Ba(OH)_2$ provide 2 mol OH^- per mole, and the keys for both 2019S-Q32 and 2024S-Q34 include this factor).
2. Compare $n(H^+)$ and $n(OH^-)$ using 1:1 neutralisation to determine the excess or limiting species (key for 2019S-Q32: H^+ is the limiting reagent).
3. If OH^- is in excess: $c(OH^-) = n(excess)/V$ (total volume); $[H^+] = Kw/[OH^-]$; $pH = -\log[H^+]$. If H^+ is in excess: $c(H^+) = n(excess)/V$; $pH = -\log[H^+]$.
4. Pure water: $[H^+] = [OH^-] = \sqrt{Kw}$, which depends on temperature. The pH value increases as temperature decreases, but pure water is still neutral (2017S-Q31(d)–(e)).
5. From pH, find $[H^+] = 10^{-pH}$; from $[H^+]$, find $[OH^-] = Kw/[H^+]$.

Mark-earning points (typical wording in the key): ✓ 2024S-Q34 (8 marks): 1 mark each for $n(H^+)$ and $n(OH^-)$ (OH^- multiplied by 2); determining that OH^- is in excess; $n(OH^-)excess = 0.00762$ mol; $c = 0.08659$ mol L^{-1}; finding $[H^+]$ from Kw; pH = 12.9. ✓ 2019S-Q32 (9 marks): $n(H^+)$, $n(OH^-)$ (multiply by 2), determining the limiting reagent, the excess species, $[OH^-]$, $[H^+]$, pH = 12.1. ✓ 2017S-Q31(d) (2 marks): from $Kw = 5.48 \times 10^{-14}$ at 50 °C, find $[H^+] = \sqrt{Kw}$, pH = 6.63. ✓ 2017S-Q31(e) (2 marks): in pure water $[H^+] = [OH^-]$, so it is neutral at 50 °C, and give the reason. ✓ 2021S-Q29(b) (2 marks): $[H^+] = [OH^-] = \sqrt{(5.13 \times 10^{-15})} = 7.16 \times 10^{-8}$ mol L^{-1}. ✓ 2016E-Q42(f) (6 marks): show every calculation step, and finally judge whether it is acid rain; no working, no marks.

Common errors (exam reports): The exam reports do not comment specifically on this question type. The following common errors are summarised from the key:

- Ignoring the 2× OH^- from $Ca(OH)_2$ or $Ba(OH)_2$.
- Forgetting to use the total volume after mixing (the key for 2024S-Q34 uses 0.088 L, i.e. 38.00 mL + 50.00 mL).
- Still using Kw = 1.00 × 10^{-14} for calculations at high temperature.
- Interpreting a pH greater than 7 as "water being basic at high temperature"; pure water satisfies $[H^+] = [OH^-]$ at any temperature.
- Writing only the answer with no intermediate steps (2016E-Q42(f)).
- Incorrect significant figures (QNT7) and units.
- MC questions cover converting between pH and $[H^+]$, solutions with pH below zero, and the self-ionisation of pure water (e.g. 2019M-Q16, 2022M-Q12, 2023M-Q19); pH can be negative, and $[H^+] = 10^{\wedge}(-pH)$ still applies.

Past questions: 2016E-Q42(f), 2016E-Q42(g), 2017S-Q29(c), 2017S-Q31(d), 2017S-Q31(e), 2018S-Q29(c), 2019M-Q16, 2019S-Q32, 2020M-Q5, 2020S-Q30(b), 2021M-Q14, 2021S-Q29(b), 2022M-Q12, 2023M-Q19, 2023S-Q34(b), 2024S-Q34, 2025E-Q36(a)

Pattern 6: Indicators and end points

Identify: "Select the most suitable indicator from the table" "Identify the most appropriate indicator for this titration and justify your choice" "Would the calculated concentration be too large or too small?". The question stem usually gives the indicator and its colour-change pH range (or colour change).

Steps

1. Determine the equivalence-point pH: strong acid with strong base is about 7; weak acid with strong base (forming a weakly basic salt) is basic, about 9; determined by salt hydrolysis (Pattern 3).
2. Find an indicator whose colour-change range covers, or is closest to, the equivalence-point pH.
3. Write the reason for the choice: the indicator's end point (the point of colour change) should be as close as possible to the equivalence point (2021S-Q28(a)).
4. State the colour change: the colour from the acidic side to the basic side (e.g. phenolphthalein colourless → pink, methyl orange red → yellow).
5. If the indicator's end point deviates from the equivalence point, decide whether the titration reading is too high or too low, and determine the change in the calculated concentration (2019S-Q35(b)).
6. The number of equivalence points for a polyprotic acid depends on the acid: H_2SO_3 has two (the context of 2019S-Q35(b)); H_3PO_4 has three, and the pH of the third equivalence point is outside phenolphthalein's colour-change range (mentioned in the key for 2018S-Q27(d)). If the indicator only changes colour at the first equivalence point, then titration only reaches the first end point; the key accepts "too low", or "unchanged" when accompanied by the reason "the first equivalence point has been reached and the calculation needs correction".

Marking points (typical key wording): ✓ 2016S-Q34(b) (4 marks): choose Bromocresol Green (colour-change range 3.8–5.4, covering the equivalence point at about pH 4.8); state that other indicators change colour before or after the equivalence point, causing systematic error. ✓ 2018S-Q27(d) (5 marks): key is phenolphthalein. The reason must include five valid marking points and one equation: at the equivalence point the solution contains PO_4^{3-} ($3OH^- + H_3PO_4 \rightarrow PO_4^{3-} + 3H_2O$); PO_4^{3-} hydrolyses to produce OH^- ($PO_4^{3-} + H_2O \rightleftharpoons HPO_4^{2-} + OH^-$); the equivalence point is weakly basic (pH about 9); the indicator's colour-change pH is close to the equivalence-point pH; Na^+ is a neutral ion and does not affect pH. If the hydrolysis equation is missing, maximum 4 marks. Methyl orange is also acceptable, but it must be based on the first equivalence point (pH about 4.7) and include the equation. ✓ 2020E-Q37(d) (5 marks): choose phenolphthalein; write the equation $OH^- + H_3PO_4 \rightarrow PO_4^{3-} + H_2O$ or the phosphate hydrolysis equation; the equivalence-point pH is about 9, close to the indicator's colour-change point. The key notes: without a hydrolysis or reaction equation, maximum 4 marks. ✓ 2021S-Q28(a) (3 marks): the equivalence-point pH is about 9; cresolphthalein

(pH 8.2–9.8) is the only indicator that changes colour at this pH. ✓ 2023S-Q33(b) (5 marks): choose phenolphthalein; at the equivalence point $[OH^-] > [H^+]$; the indicator changes colour in the basic range; write the NH_3 hydrolysis equation. ✓ 2017E-Q37(d) (4 marks): the key lists both bromothymol blue and phenolphthalein; the reason is that the equivalence-point pH = 7 (strong acid–strong base), and the colour-change points of these two indicators are close to 7. Note: phenolphthalein's colour-change range is 8.3–10.0, which is quite far from 7. The key accepts both; answers should justify the choice by how close the colour-change range is to the equivalence point, not by name alone.

Common errors (exam reports): The 2022 report states that titration calculations and practical techniques (2022E-Q38) were challenging for many candidates (the indicator selection in 2022E-Q38(c) belongs to this pattern). The following common errors are summarised from the key:

- Choosing an indicator only because its colour change is obvious, without comparing the colour-change range with the equivalence point.
- Confusing the equivalence point with the end point, and not stating that the indicator's colour-change point should be close to the equivalence point.
- Considering only one equivalence point for a polyprotic acid: in 2019S-Q35(b), when titration only reaches the first equivalence point, the key accepts "too low" (with the reason that a smaller amount of NaOH was added), or "unchanged" when accompanied by the reason "the first equivalence point has been reached and the calculation needs correction".
- Writing only "changes colour" for the colour change, without giving the starting and final colours.
- MC questions test the nature and choice of indicators (e.g. 2019M-Q1, 2020M-Q17, 2022M-Q1); an indicator is a weak acid or base, and its colour-change range should lie within the titration jump; the equivalence point and end point are not the same (2016M-Q12).

Past questions: 2016M-Q12, 2016S-Q34(b), 2017M-Q8, 2017E-Q37(d), 2018M-Q1, 2018S-Q27(d), 2019M-Q1, 2019S-Q35(b), 2020M-Q17, 2020E-Q37(d), 2021S-Q28(a), 2022M-Q1, 2022E-Q38(c), 2023S-Q33(b), 2024M-Q6

Pattern 7: Titration techniques and apparatus

Identify: "Outline two reasons why sodium hydroxide is not a suitable primary standard" "state with what ... should be rinsed" "list three properties of a primary standard" "choose the most suitable piece of equipment" "predict the effect ... on the calculated concentration".

Steps

1. Rules for rinsing apparatus: rinse the burette with the solution it will contain (standard solution or analyte solution, depending on which is being placed in it); rinse the conical flask with distilled water (do not rinse with the analyte solution, otherwise it increases the amount of analyte); rinse the pipette with the solution to be pipetted (to ensure the amount of substance pipetted is accurate). Rinse the volumetric flask with distilled water (2019E-Q39(b)).
2. Properties of a primary standard: high purity, known purity, stable (not hygroscopic, not deliquescent, does not react with CO_2 or O_2), soluble, and has a relatively large molar mass (the key for 2021E-Q35(a) accepts any two).
3. Reasons NaOH is not a primary standard: hygroscopic, deliquescent, absorbs CO_2 to form Na_2CO_3, unknown purity (keys for 2018S-Q27(c) and 2016E-Q42(b)).
4. Effect of errors on results: first determine the change in the "amount used", then determine the direction of change in the calculated concentration. For example, residual water in the pipette dilutes the analyte solution, so less standard solution is required, and the calculated concentration is too low (2018E-Q40(d), 2019S-Q35(b)).
5. Apparatus selection: use a volumetric flask for an accurate 100.0 mL solution; use a pipette for a 20.0 mL aliquot; use a burette for titration (2021E-Q35(b)).

Marking points (typical key wording): ✓ 2016E-Q42(b) (2 marks): 1 mark for each reason (maximum 2 marks). ✓ 2018S-Q27(c) (2 marks): any two reasons, such as hygroscopic, deliquescent, absorbs CO_2, unknown purity. ✓ 2021E-Q35(a) (2 marks): any two relevant properties. ✓ 2022E-Q38(b) (3 marks): list three properties of a

primary standard (two listed = 2 marks, one = 1 mark). ✓ 2019E-Q39(b) (3 marks): rinse the 5.00 mL pipette with the concentrated herbicide; rinse the 20.00 mL pipette with the diluted herbicide; rinse the 250.0 mL volumetric flask with distilled water. ✓ 2019S-Q35(b) (4 marks): error I (residual distilled water dilutes the acid being tested) → calculated concentration too low (1 mark each for direction of change and reason); error II (titrating only to the first equivalence point) → too low (less NaOH added), or unchanged when accompanied by the reason "the first equivalence point has been reached and the calculation needs correction" (1 mark each). ✓ 2025E-Q38(e) (2 marks): rinse the conical flask with distilled water; the reason is that this must not change the amount of substance of IBA pipetted (do not introduce other substances that would react with IBA).

Common errors (exam reports): The 2022 report states that titration calculations and practical techniques (2022E-Q38) were challenging for many candidates. The following common errors are summarised from the key:

- Rinsing the burette with distilled water (it should be rinsed with the solution it will contain).
- Giving "relative molecular mass is too small" as the only reason NaOH is not a primary standard, even though the question stem already gives this (the 2018S-Q27(c) stem uses "Other than …").
- Stating the direction of the error without a reason (you need to explain dilution, a change in amount used, or the indicator's end point).
- Writing only "high purity" when listing properties of a primary standard, without giving other properties.
- MC questions ask which technique is least likely to cause error, or which final rinse solution is used for the burette and pipette (e.g. 2016M-Q7, 2020M-Q25, 2022M-Q9); both are rinsed with the solution they will hold, and readings are taken at the bottom of the meniscus.

Past questions: 2016M-Q7, 2016E-Q42(b), 2016E-Q42(d), 2018M-Q12, 2018S-Q27(c), 2018E-Q40(d), 2019S-Q35(b), 2019E-Q39(b), 2020M-Q25, 2021E-Q35(a), 2021E-Q35(b), 2022M-Q9, 2022E-Q38(b), 2023E-Q36(a), 2024M-Q7, 2025E-Q38(e)

Pattern 8: Titration curves

Recognition: "Label the equivalence point on the titration curve using an arrow and record the pH" "circle the region on the titration curve where buffering occurs" "give two reasons … making reference to the titration curve" .

Steps

1. Read the initial pH to determine whether the acid or base is strong or weak (the starting pH is related to concentration).
2. Find the sharp change (steep pH change) region; the equivalence point is at the midpoint of the sharp change (the middle of the vertical section).
3. Compare titration curves for different combinations: strong acid–strong base has an equivalence point of about 7; weak acid–strong base has an equivalence point greater than 7; a diprotic weak acid (such as H_2SO_3) shows two sharp changes; for the strong diprotic acid H_2SO_4, the first step is almost complete ionisation, so the curve has only one plateau and one end point (as stated in the key for 2019S-Q35(a)(ii)).
4. Buffer region: in the region where the weak acid and its conjugate base coexist, the curve is relatively flat (2021S-Q28(bi)); circle this flat region, not the area near the equivalence point.

Marking points (typical key wording): ✓ 2016S-Q34(a) (2 marks): Use an arrow to mark the equivalence point and record the pH as about 4.8 (the key accepts 4.3–5.3). ✓ 2021S-Q28(bi) (1 mark): Circle the flat buffer region on the curve. ✓ 2019S-Q35(a)(ii) (2 marks): 1 mark for each reason. The key accepts: H_2SO_4 is a strong acid with an initial pH less than 1, whereas H_2SO_3 is a weak acid with an initial pH greater than 1; or H_2SO_4 has only one sharp change and one end point, whereas H_2SO_3 has two equivalence points at about pH 4.2 and 9.6, indicating it is a weak acid. (The key for (a)(i) is sulfurous acid, 1 mark.)

Common errors (exam reports): The exam reports do not have specific comments on this pattern. The following common errors are summarised from the key:

- Marking the equivalence point outside the sharp change, or marking only the pH and not the position.
- Circling the buffer region near the equivalence point instead of the flat region.
- Considering only one sharp change for a polyprotic acid.

Past questions: 2016S-Q34(a), 2019S-Q35(a)(ii), 2021S-Q28(a), 2021S-Q28(b)(i)

Pattern 9: Non-redox reaction equations and observations ★

Recognition: "Write a balanced chemical equation, showing all state symbols" "Describe the observations" "Predict all visible changes" "Write the balanced ionic equation" "combine equations into an overall equation"; reaction types include acid with carbonate, acid with metal oxide or hydroxide, precipitation, and thermal decomposition (such as $Cu(OH)_2$).

Steps

1. Identify the reaction type (acid–base neutralisation, precipitation, gas release, decomposition). Use the solubility rules in the data booklet to decide whether a precipitate forms, and use the colour table to decide the colour of the precipitate or solution (such as $CuCO_3$ green, CuO black).
2. Write the species: (s) for solid, (ℓ) for liquid, (g) for gas and (aq) for solution; write weak acids as molecular formulae (such as $H_2CO_3(aq)$).
3. Balance atoms and charge (ionic equations). If the reaction has "CO_2 escaping", write $CO_2(g)$ as a product (2018S-Q26(a) accepts either H_2CO_3 or CO_2).
4. Describe observations: solid dissolves or disappears, colourless odourless bubbles are produced (effervescence), initial and final colour of the solution, and colour and state of the precipitate.
5. Combine multi-step reactions: remove intermediate species (such as the relationship between CO_2 and HCO_3^-, 2019E-Q36(c)).
6. Precipitate weighing question: precipitate → filter → wash → dry → weigh, and write the equation for the precipitation reaction (2019S-Q30(a)).
7. Calculate precipitate mass and amount of substance: use the stoichiometric ratio (such as $3Cd^{2+}$: $2PO_4^{3-}$), and take the upper limit of the error range to ensure all ions are precipitated (2018E-Q36(c)).

Marking points (typical key wording): ✓ 2018S-Q26(a) (3 marks): correct species (1 mark), balanced (1 mark), state symbols (1 mark). ✓ 2018S-Q26(b) (3 marks): any three observations (such as blue solid disappears, green precipitate or solid forms, solution is colourless after the reaction is complete, bubbles). ✓ 2018S-Q26(c) (2 marks): any two different observations (such as no solid forms, solution is finally blue, reaction is faster, no gas is produced). ✓ 2019S-Q26(a) (4 marks): test tube I contains a salmon pink (copper-coloured) solid and a colourless solution; test tube II contains a black solid and a blue (green) solution. ✓ 2022E-Q37(b) (2 marks): describe all observations, including colour changes, reactants and products. ✓ 2022E-Q37(c) (3 marks): equation for the reaction of CuO with dilute sulfuric acid, including correct species, balancing and state symbols. ✓ 2023S-Q30(b) (4 marks): $Ca(HCO_3)_2(s) + 2H^+(aq) \rightarrow Ca^{2+}(aq) + 2H_2O(\ell) + 2CO_2(g)$; observations are white solid dissolves, colourless solution, bubbles. ✓ 2023S-Q33(a) (2 marks): $OH^-(aq) + NH_4^+(aq) \rightarrow NH_3(aq) + H_2O(\ell)$ (ionic equation, correct species and balancing). ✓ 2018E-Q36(c) (8 marks): calculate the minimum mass of sodium phosphate (upper limit of error, mole ratio, molar mass, significant figures).

Common errors (exam reports): The exam reports do not have specific comments on this pattern. The following common errors are summarised from the key:

- State symbols omitted or written incorrectly ($I_2(s)$ and $I_2(aq)$ are different; see the key for RX3).
- Equation not balanced (even if the species are correct, only some marks are awarded).
- Observations only say "colour change", without the initial and final colours.
- Incorrect relationship between H_2CO_3 and CO_2, or missing CO_2 as a product.
- In 2018S-Q26(c), writing the reaction of HCl with $Cu(OH)_2$ as producing a gas, when in fact no gas is released.
- MC questions use solubility rules to decide whether ions can coexist or precipitate (e.g. 2023M-Q3);

ions that form insoluble chlorides, such as Ag^+, are unlikely to be present in large amounts in seawater.

Past questions: 2016S-Q27(a)(ii), 2016S-Q33(a), 2017E-Q37(a), 2018S-Q26(a), 2018S-Q26(b), 2018S-Q26(c), 2018E-Q36(c), 2019S-Q26(a), 2019S-Q26(b), 2019S-Q26(c), 2019S-Q27(b), 2019S-Q30(a), 2019E-Q36(c), 2019E-Q37(b), 2020E-Q37(d), 2022E-Q37(b), 2022E-Q37(c), 2022E-Q37(d), 2023M-Q3, 2023M-Q9, 2023S-Q27(a), 2023S-Q30(b), 2023S-Q33(a), 2025E-Q35(c)